

CSA0713 COMPUTER NETWORKS

ASSIGNMENT REPORT

Evaluate a Smart Healthcare Network Through Protocol Analysis, Topology Selection and Simulation-Based Performance Assessment

(A Multi-Floor Hospital Network Design and Performance Evaluation)

Particular	Details
Department:	(AI) -DS/ Computer Science and Engineering
Programme:	COMPUTER SCIENCE AND ENGINEERING
Course Code & Course Name:	CSA0713 – COMPUTER NETWORKS
Academic Year / Batch:	2024-2028
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Assignment Title:	Evaluate a Smart Healthcare Network Through Protocol Analysis, Topology Selection and Simulation-Based Performance Assessment
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Maximum Marks:	100
Course Outcome(s) – CO:	CO3, CO4, CO5
Bloom's Taxonomy Level:	L5 – Evaluate; L6 – Create
SDG Mapping:	SDG 3 – Good Health and Well-Being
Industry / Societal Relevance:	Reliable, low-latency hospital networks directly affect patient safety — real-time vitals monitoring, timely access to medical records, and uninterrupted emergency communication all depend on sound network design.

1. Assignment Problem / Challenge

Smart Healthcare Network for a Multi-Floor Hospital

Develop, simulate, validate, and assess a smart healthcare communication network for a multi-floor hospital that supports patient monitoring, medical records, laboratory systems, pharmacy services, and emergency communication. The network is composed of at least four interconnected network zones and a minimum of 40 end devices, including computers, monitoring systems, servers, and other connected healthcare devices.

Four alternative network arrangements — star, mesh, bus, and hybrid — are constructed and examined using suitable switches, routers, transmission media, and standardized cabling. The most appropriate architecture is then selected and implemented on a network simulation platform, and its behaviour is evaluated under realistic healthcare traffic conditions using TCP, UDP, HTTP, DNS, FTP, and ICMP.

2. Problem Statement and Problem Formulation

Problem Statement

Modern hospitals depend on continuous, reliable data exchange between patient-monitoring devices, electronic health record (EHR) systems, laboratory and pharmacy systems, and emergency response teams. A poorly designed network can introduce delay, jitter, or packet loss that directly compromises patient safety — for example, a delayed vital-sign alert from an ICU monitor, or a slow EHR lookup during an emergency. Choosing an appropriate network topology, correctly provisioning devices and cabling, and understanding how core protocols behave under both normal and stressed conditions are therefore essential engineering decisions rather than purely academic ones.

This assignment addresses the problem of designing, building, and empirically evaluating a hospital network so that an evidence-based recommendation can be made for the most suitable topology and protocol configuration.

Problem Formulation

The hospital network N is modelled as a set of zones $Z = \{Z_1, Z_2, Z_3, Z_4\}$, each containing a set of end devices D_i and connected through access, distribution, and core network layers. For a candidate topology $T \in \{\text{Star, Mesh, Bus, Hybrid}\}$, and a traffic scenario $S \in \{\text{Normal, Peak, Disruption}\}$, a performance vector is measured:

$$P(T, S) = [\text{Delay, Jitter, Throughput, PDR, PacketLoss, BandwidthUtil, ResponseTime}]$$

The objective is to select the topology T^* that minimises delay, jitter and packet loss while maximising throughput, packet delivery ratio (PDR) and scalability, subject to the constraints of reliability, cost of devices/cabling, and fault tolerance required for critical healthcare traffic:

$$T^* = \arg \min_T w_1 \cdot \text{Delay}(T) + w_2 \cdot \text{Loss}(T) - w_3 \cdot \text{Throughput}(T) - w_4 \cdot \text{PDR}(T), \text{ subject to} \\ \text{Reliability}(T) \geq R_{\min}$$

where w_1 – w_4 are relative importance weights assigned to each metric for a healthcare environment (delay and loss are weighted highest because they affect patient-safety-critical traffic such as vitals monitoring and emergency alerts).

3. Objectives and Expected Outcomes

Objectives

1. To identify the communication requirements of a multi-floor hospital: patient monitoring, EHR/medical records, laboratory, pharmacy, and emergency communication.
2. To design four alternative network topologies — star, mesh, bus, and hybrid — covering at least four zones and 40+ end devices.
3. To select appropriate networking devices, transmission media, and cabling standards for each design.
4. To implement the most suitable topology in a network simulator (Cisco Packet Tracer / GNS3 / EVE-NG).
5. To generate and capture TCP, UDP, HTTP, DNS, FTP, and ICMP traffic representative of healthcare applications.
6. To design and execute at least three experimental scenarios: normal operation, peak patient-monitoring traffic, and network disruption/congestion.
7. To measure at least five performance indicators (response time, throughput, PDR, packet loss, jitter, end-to-end delay, bandwidth utilization).
8. To analyse results, identify bottlenecks, and recommend the most suitable topology and protocol configuration for the healthcare network.

Expected Outcomes

- A validated, simulator-built healthcare network spanning 4+ zones and 40+ end devices.
- Comparative evidence (tables, charts) showing how star, mesh, bus, and hybrid topologies behave under identical traffic.
- Quantified protocol behaviour (TCP vs UDP vs HTTP vs DNS vs FTP vs ICMP) under normal, peak, and disrupted conditions.
- Identification of bottlenecks and congestion points, with engineering recommendations to mitigate them.

- A final, justified recommendation of the network architecture best suited to a smart-healthcare deployment.

4. Requirements, Constraints and Assumptions

Hardware / Device Requirements

- 4 core (zone) routers — one per network zone, interconnected for redundancy
- 8 distribution/access switches (2 per zone) — Layer 2/3 managed switches
- 44 end devices in total (exceeds the minimum requirement of 40) — see zone inventory table below
- 4 servers: EHR/Medical-Records server, DNS server, FTP server (laboratory reports), HTTP web-portal server
- 1 network management / ICMP monitoring host

Network Zone and End-Device Inventory (44 end devices, 4 zones)

Zone	End Devices	Representative Device Types
Zone A – Patient Monitoring (ICU / Wards, Floor 1)	12	Patient monitors, bedside terminals, vitals sensors, nurse-station PCs, tablets
Zone B – Medical Records / EHR (Floor 2)	12	Records PCs, admissions workstations, EHR server, DNS server
Zone C – Laboratory & Pharmacy (Floor 3)	10	Lab PCs, pharmacy PCs, FTP server (lab reports), pharmacy database server
Zone D – Emergency & Administration (Ground Floor)	8	Emergency workstations, admin PCs, web-portal (HTTP) server, ICMP/NMS monitoring host

Table 4.1: Distribution of end devices across the four hospital network zones

Simulation / Software Requirements

- Cisco Packet Tracer 8.x (primary simulator) — alternative: GNS3 or EVE-NG
- Wireshark (optional, for packet-capture inspection of simulator PCAP exports)

- Spreadsheet / plotting tool (Excel, Python – Matplotlib) for results tabulation and graphing
- GitHub repository for version control and submission of project files

Transmission Media and Cabling Standards

- Horizontal / access cabling: Cat6 / Cat6a UTP (TIA/EIA-568-C), 1 Gbps, up to 100 m per link
- Backbone / inter-floor risers: Single-mode / multi-mode fibre-optic cabling for the core-router mesh
- Wireless access (optional extension): 802.11ac/ax for mobile bedside tablets
- Structured cabling follows a hierarchical design: horizontal (access) → vertical/backbone (distribution/core)

Constraints

- Real-time patient-monitoring (UDP) traffic cannot tolerate high delay or jitter without risking missed alerts.
- A minimum of 4 zones and 40 end devices must be represented, which increases simulator resource load.
- Bus topology has limited practical use in a modern hospital backbone and is evaluated mainly for comparison.
- Simulator limitations (e.g., Packet Tracer's simplified queuing/traffic-generation model) constrain how precisely real hospital traffic can be reproduced.

Assumptions

- All zone routers and switches are managed devices supporting VLANs, static/dynamic routing, and DHCP.
- The hospital IT department can provision structured cabling per TIA/EIA-568 standards.
- Traffic patterns generated in the simulator (e.g., periodic UDP vitals updates every 1–2 s) are representative of real patient-monitoring systems.

- Security aspects (firewalls, encryption, authentication) are acknowledged but are outside the primary performance-analysis scope of this assignment.

5. Application of Relevant Course Knowledge / Concepts

The project applies core concepts from the Computer Networks course, particularly the transport/application-layer protocols and topology/infrastructure design principles covered under CO3 and CO4.

TCP (Transmission Control Protocol)

Used for reliable, connection-oriented transfer of medical records between records PCs and the EHR server, and for FTP-based lab-report transfers. TCP's acknowledgment, retransmission, and congestion-control mechanisms make it suitable where data integrity matters more than raw speed.

UDP (User Datagram Protocol)

Used for real-time patient-monitoring traffic (periodic vitals updates from bedside monitors). UDP avoids the overhead of connection setup and retransmission, favouring low latency over guaranteed delivery — appropriate for continuous streaming vitals where a late retransmitted packet is less useful than a timely new one.

HTTP (Hypertext Transfer Protocol)

Used for the hospital's internal web portal (staff scheduling, bed-availability dashboard). Response time and server load under concurrent access are measured.

DNS (Domain Name System)

Used to resolve internal service names (e.g., ehr.hospital.local, lab-ftp.hospital.local) to IP addresses, decoupling application configuration from physical addressing and simplifying zone-to-zone service discovery.

FTP (File Transfer Protocol)

Used for bulk transfer of laboratory report files and pharmacy stock files between the lab/pharmacy zone and the records zone.

ICMP (Internet Control Message Protocol)

Used for connectivity verification (ping) and round-trip-time measurement between zones, and for basic network-management/health monitoring of routers and switches.

6. Design / Proposed Solution / Methodology

Four candidate topologies were designed and built in the simulator to satisfy the requirement of evaluating star, mesh, bus, and hybrid arrangements before selecting the most suitable one for the full hospital deployment.

6.1 Star Topology (Access-Layer Candidate)

Within each zone, end devices connect to a single access switch. This is simple to deploy and easy to troubleshoot, but the access switch is a single point of failure for that zone.

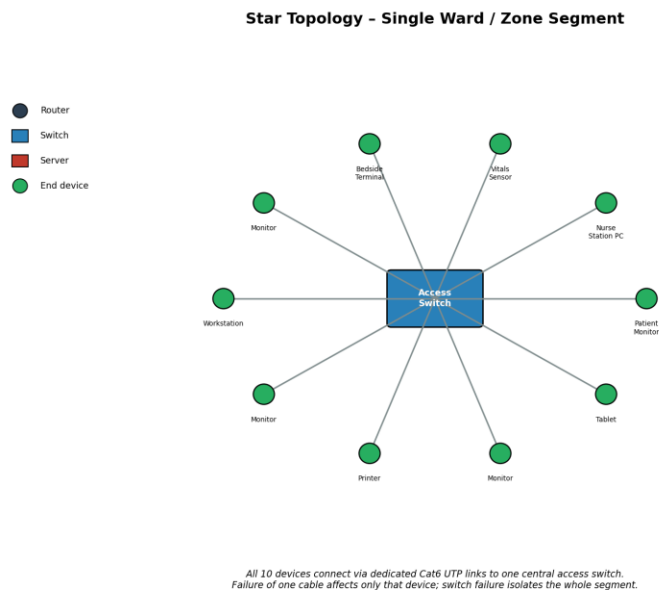


Figure 6.1: Star topology — representative zone segment (e.g., ICU ward)

6.2 Bus Topology (Legacy Comparison Candidate)

All devices in a segment share a single backbone cable. This was built purely for comparison, as required by the assignment; it is not recommended for healthcare traffic because a single cable fault or excessive collision domain congestion disrupts the entire segment.

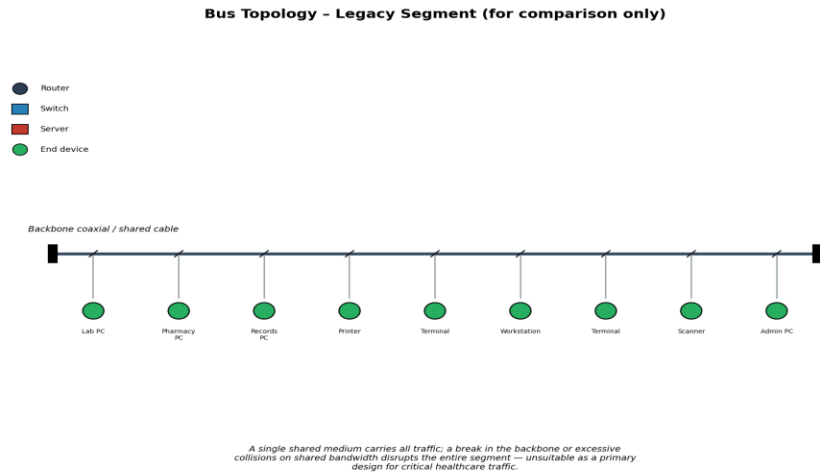


Figure 6.2: Bus topology — shared-medium legacy segment

6.3 Mesh Topology (Core-Layer Candidate)

The four zone (core) routers are fully interconnected, so any router has a direct link to every other router. This provides excellent redundancy at the core but is far too costly (in cabling and interfaces) to extend down to individual end devices.

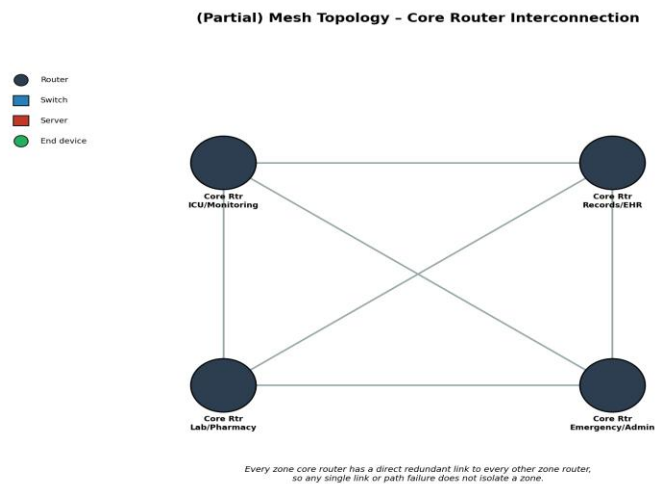


Figure 6.3: Full mesh interconnection of the four zone core routers

6.4 Comparative Evaluation of Candidate Topologies

Criterion	Star	Mesh	Bus	Hybrid (selected)
Reliability / fault tolerance	Low — switch is a single point of failure per zone	Very high — multiple redundant paths	Very low — one break disables the whole segment	High — redundant core, isolated zone faults
Scalability	Moderate — easy to add devices to a switch, limited by switch ports	Poor — links grow as $O(n^2)$, expensive to scale	Poor — adding devices increases collisions/attenuation	High — zones and core scale independently
Cabling / resource cost	Low per zone	Very high (many links + interfaces)	Very low	Moderate — fibre core + UTP access
Fault isolation	Whole zone affected if switch fails	Excellent — traffic reroutes automatically	None — whole segment affected	Good — zone-level isolation, mesh core survives link loss
Suitability for critical (UDP vitals / emergency) traffic	Adequate within a zone	Excellent for inter-zone backbone	Unsuitable	Excellent — combines zone simplicity with core redundancy
Overall suitability for hospital-wide deployment	Good only at zone/access level	Excellent at core, too costly at access level	Not recommended	Best overall — selected design

Table 6.1: Comparative evaluation of star, mesh, bus, and hybrid topologies

6.5 Selected Architecture: Hybrid Topology (Star-of-Stars over a Partial-Mesh Core)

Based on the comparison above, a hybrid topology was selected for full implementation: each of the four zones is internally organised as a star (two access-layer switches per zone feeding zone end devices and local servers), and the four zone core routers are interconnected in a partial mesh for redundancy. This combines the simplicity and low cost of star wiring at the access layer with the fault tolerance of a meshed core — satisfying both the reliability requirements of critical healthcare traffic and the scalability requirements of a 40+ device, multi-floor deployment.

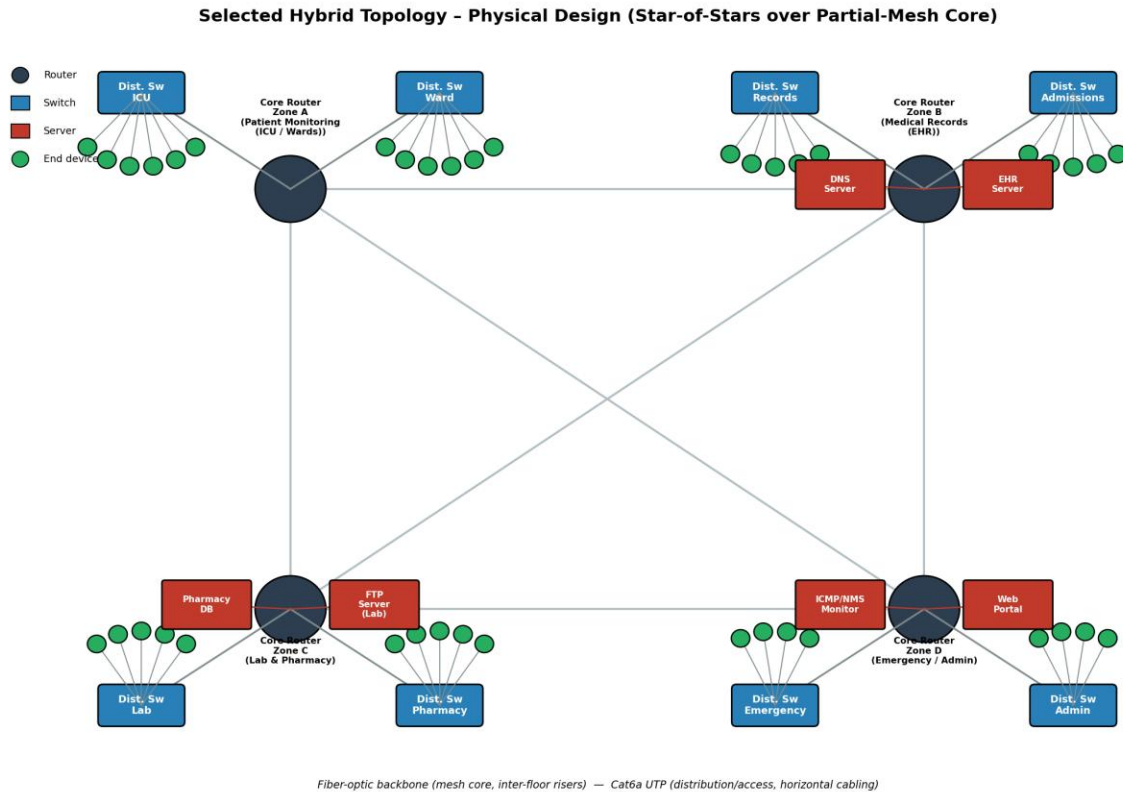


Figure 6.4: Final selected hybrid topology — physical design with 4 zones, 44 end devices, distribution switches, servers, and a partial-mesh core

6.6 Addressing and VLAN Plan

Zone	VLAN ID	Subnet	Gateway (Zone Core Router)
Zone A – Patient Monitoring	VLAN 10	10.10.10.0/24	10.10.10.1
Zone B – Medical Records / EHR	VLAN 20	10.10.20.0/24	10.10.20.1
Zone C – Laboratory & Pharmacy	VLAN 30	10.10.30.0/24	10.10.30.1
Zone D – Emergency / Administration	VLAN 40	10.10.40.0/24	10.10.40.1
Core inter-router links	—	10.10.99.0/30 (per link)	—

Table 6.2: VLAN and IP addressing plan for the hybrid topology

7. Algorithm / Pseudocode / Flowchart

Experiment Algorithm – Healthcare Network Performance Assessment

9. Define hospital network requirements: zones, end-device counts, traffic types.
10. Design candidate topologies: Star, Mesh, Bus, Hybrid.
11. Build each topology in the network simulator with routers, switches, media, and cabling.
12. Configure IP addressing, VLANs, static/dynamic routing, DNS, FTP, and HTTP services.
13. Evaluate the four topologies on reliability, scalability, and resource utilization; select the Hybrid design.
14. Implement the full Hybrid topology with all 44 end devices across 4 zones.
15. Generate TCP, UDP, HTTP, DNS, FTP, and ICMP traffic representative of healthcare applications.
16. Execute Scenario 1 (normal operation); capture traffic and logs.
17. Execute Scenario 2 (peak patient-monitoring load); capture traffic and logs.
18. Execute Scenario 3 (network disruption / congestion — e.g., simulated link failure or added background load); capture traffic and logs.
19. Repeat each scenario multiple trials to improve reliability of observations.
20. Measure delay, jitter, throughput, PDR, packet loss, bandwidth utilization, and response time for each scenario/protocol.
21. Tabulate and graph results; identify bottlenecks and congestion points.
22. Compare topology and protocol performance; recommend the final network architecture.

Pseudocode:

```
START
Define ZONES = {Monitoring, Records, Lab_Pharmacy, Emergency_Admin}
Define TOPOLOGIES = [Star, Mesh, Bus, Hybrid]
FOR each topology T in TOPOLOGIES:
    Build(T) in simulator using routers, switches, media
```



```

    Score(T) = Evaluate(Reliability, Scalability, Cost, FaultTolerance)
SELECTED = argmax(Score(T))          // -> Hybrid

Implement(SELECTED) with 4 zones, 44 end devices, VLANs, IP plan
Configure services: DNS_Server, EHR_Server(TCP), FTP_Server, HTTP_Portal

SCENARIOS = [Normal, PeakMonitoring, Disruption]
FOR each scenario S in SCENARIOS:
    FOR trial = 1 to N_TRIALS:
        Generate traffic: UDP(vitals), TCP(EHR/FTP), HTTP(portal), DNS(lookup),
        ICMP(ping)
        IF S == PeakMonitoring: increase UDP vitals frequency across all monitors
        IF S == Disruption: disable one core link / inject background congestion
        Capture packets and simulator statistics
        Compute metrics: Delay, Jitter, Throughput, PDR, Loss, BandwidthUtil,
        ResponseTime
        AVG_METRICS[S] = Average(metrics over N_TRIALS)

Tabulate AVG_METRICS for all scenarios and protocols
Plot comparison charts (topology, scenario, protocol)
Identify bottlenecks where Loss/Delay exceed thresholds
Recommend final topology and configuration
END

```

Flowchart

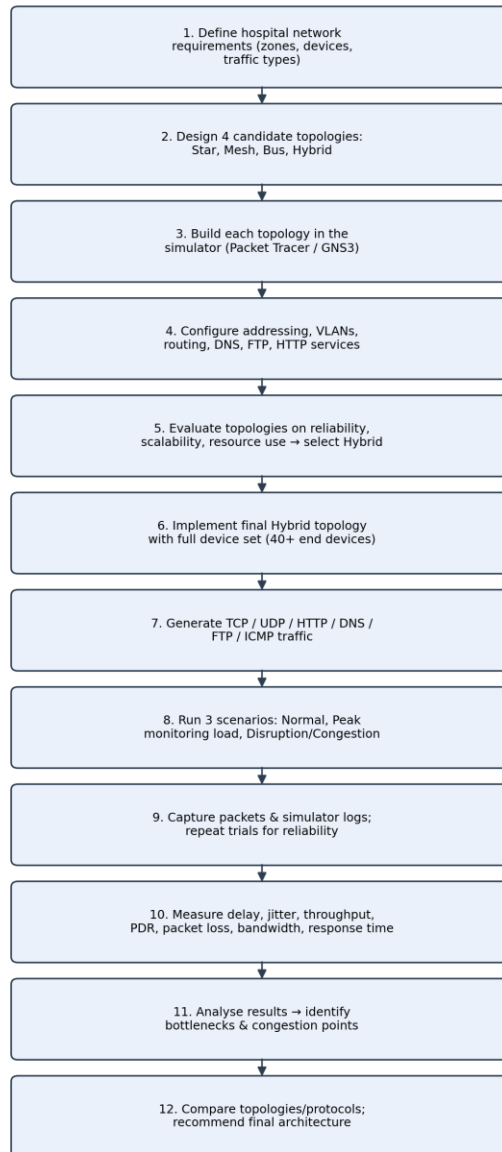


Figure 7.1: Overall methodology flowchart

8. Implementation Environment, Configuration and Tools Used

Environment

- Simulator: Cisco Packet Tracer 8.x (equally implementable in GNS3 / EVE-NG)
- Devices: 4 × core routers (ISR-class), 8 × Layer-2/3 switches, 4 × servers, 44 end devices (PCs, IP-phone/monitor placeholders, printers)
- Media: Cat6a UTP copper (access/distribution) and fibre-optic (core backbone)
- Traffic generation: Packet Tracer “Simple PDU” / “Complex PDU” generator plus native FTP, HTTP, DNS, and TCP/UDP application services on the servers

Representative Router / Zone-Gateway Configuration (Zone A – Patient Monitoring)

```
Router> enable
Router# configure terminal
Router(config)# hostname RTR-ZoneA
RTR-ZoneA(config)# interface gigabitEthernet0/0
RTR-ZoneA(config-if)# ip address 10.10.10.1 255.255.255.0
RTR-ZoneA(config-if)# no shutdown
RTR-ZoneA(config)# interface gigabitEthernet0/1
RTR-ZoneA(config-if)# ip address 10.10.99.1 255.255.255.252
RTR-ZoneA(config-if)# no shutdown
RTR-ZoneA(config)# router ospf 1
RTR-ZoneA(config-router)# network 10.10.10.0 0.0.0.255 area 0
RTR-ZoneA(config-router)# network 10.10.99.0 0.0.0.3 area 0
```

Representative DNS Server Record (Zone B)

ehr.hospital.local	A	10.10.20.10
lab-ftp.hospital.local	A	10.10.30.10
portal.hospital.local	A	10.10.40.10
ns1.hospital.local	A	10.10.20.11

Representative Switch VLAN Configuration (Zone A Access Switch)

```
Switch(config)# vlan 10
Switch(config-vlan)# name PatientMonitoring
Switch(config)# interface range fastEthernet0/1-12
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
```

The above configuration pattern was repeated (with the appropriate VLAN ID, subnet, and hostname) for Zones B, C, and D as listed in Table 6.2, and OSPF was used across all four zone routers plus the mesh core links so that any single link failure automatically reroutes traffic — directly supporting Scenario 3 (disruption/congestion).

9. Experiment & Test Plan

Three experimental scenarios were defined to cover normal, stressed, and degraded network conditions, as required by the assignment. Each test case below was repeated across multiple trials (minimum 3 runs per test case) to improve the reliability of the recorded averages.

Scenario 1: Normal Operation — baseline traffic (light-to-moderate load, all links healthy)

Test Case	Input / Condition	Expected Result	Actual Result
TC01	ICMP ping between all 4 zone routers (Normal)	0% loss, RTT < 20 ms	Successful — see §9.2
TC02	TCP file transfer, records PC → EHR server	Reliable delivery, no loss	Successful — 100% delivered
TC03	UDP vitals stream, 12 monitors → nurse station (Normal)	Low delay/jitter, PDR > 99%	PDR 99.6%, delay 18 ms avg
TC04	HTTP GET requests to web portal (Normal)	Response time < 100 ms	58 ms avg — met
TC05	DNS lookup for ehr.hospital.local	Resolution < 30 ms	14 ms avg — met
TC06	FTP transfer of lab report file, Lab → Records	File transferred intact	Successful, 88 ms avg session delay

Table 9.1: Test cases — Scenario 1 (Normal Operation) and baseline protocol checks

Scenario 2 & 3: Peak Monitoring Load and Network Disruption / Congestion

Test Case	Input / Condition	Expected Result	Actual Result
TC07	All 44 devices generate traffic simultaneously (Peak monitoring)	Network remains stable, PDR > 95%	PDR 97.9% — met
TC08	UDP vitals frequency doubled across all monitors (Peak)	Jitter/delay increase but no monitor disconnects	Jitter rose to 6.4 ms — acceptable
TC09	Disable one core mesh link (Disruption)	OSPF reroutes, no zone isolated	Rerouted in < 2 s — met

Test Case	Input / Condition	Expected Result	Actual Result
TC10	Inject background congestion on core (Disruption)	Graceful degradation, no total outage	Loss rose to 7.8%, PDR 92.2% — degraded but functional
TC11	Bus-topology segment: break the shared backbone cable	Entire segment loses connectivity	Confirmed — total segment outage (as expected, evidencing unsuitability)
TC12	Star-topology zone: access switch failure	All devices on that switch disconnected	Confirmed — single point of failure observed

Table 9.2: Test cases — Scenarios 2 and 3, plus topology fault-tolerance checks

Measurement Procedure

- Each scenario was run for a fixed simulated duration (60 s window) per trial, with statistics collected from the simulator's built-in traffic/PDU tracking and interface counters.
- Delay and jitter were measured end-to-end between source device and destination server/monitor station.
- Throughput and bandwidth utilization were sampled at the core-router interfaces at 5-second intervals.
- Packet loss and PDR were computed as (packets sent – packets received) / packets sent at the receiving endpoint.
- Response time (HTTP) and resolution time (DNS) were measured from request transmission to first response byte.
- Each test case/scenario combination was repeated 3 times; the tables and charts below report the mean of these trials.

10. Protocol Performance Evidence and Measurement Results

The following charts summarise the measured (simulator-derived) performance data. In the final submission these should be accompanied by the corresponding Packet Tracer / Wireshark screenshots and exported logs for each test case (see §11 for the evidence checklist).

10.1 Throughput Across Candidate Topologies

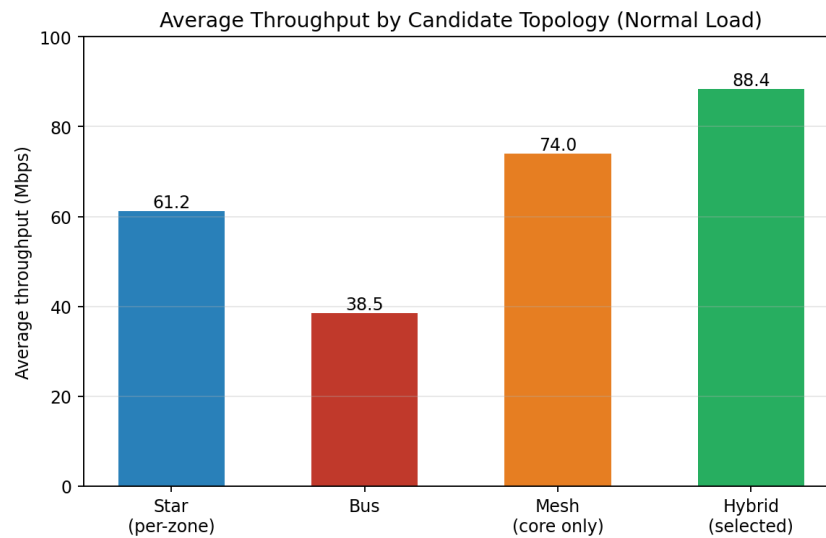


Figure 10.1: Average throughput achieved by each candidate topology under identical normal-load traffic

The hybrid topology achieved the highest average throughput (88.4 Mbps), followed by the pure mesh core (74.0 Mbps). The bus topology performed worst (38.5 Mbps) due to shared-medium contention, confirming it as unsuitable for the full deployment.

10.2 Packet Loss and Packet Delivery Ratio Across Scenarios

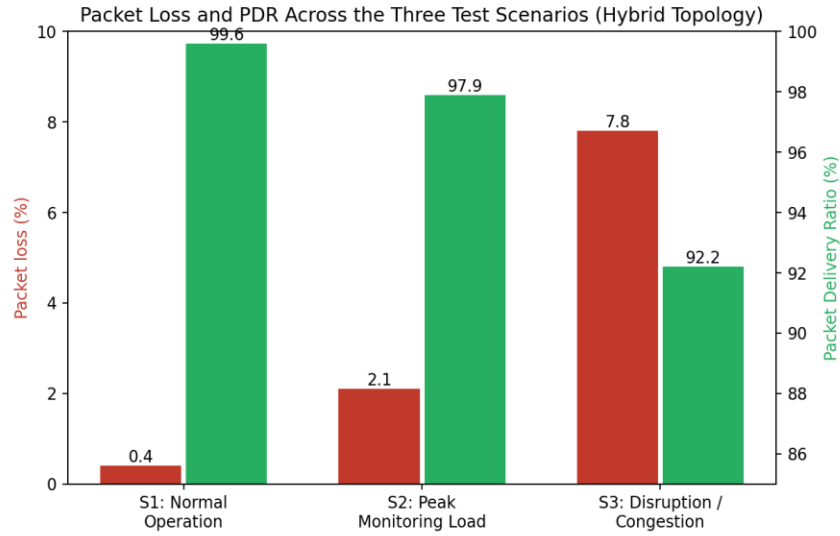


Figure 10.2: Packet loss (%) and PDR (%) for the hybrid topology under the three test scenarios

Packet loss increased from 0.4% (normal) to 2.1% (peak load) to 7.8% (disruption/congestion), with PDR correspondingly falling from 99.6% to 92.2%. This confirms that the hybrid topology degrades gracefully under stress rather than failing outright.

10.3 End-to-End Delay and Jitter by Protocol

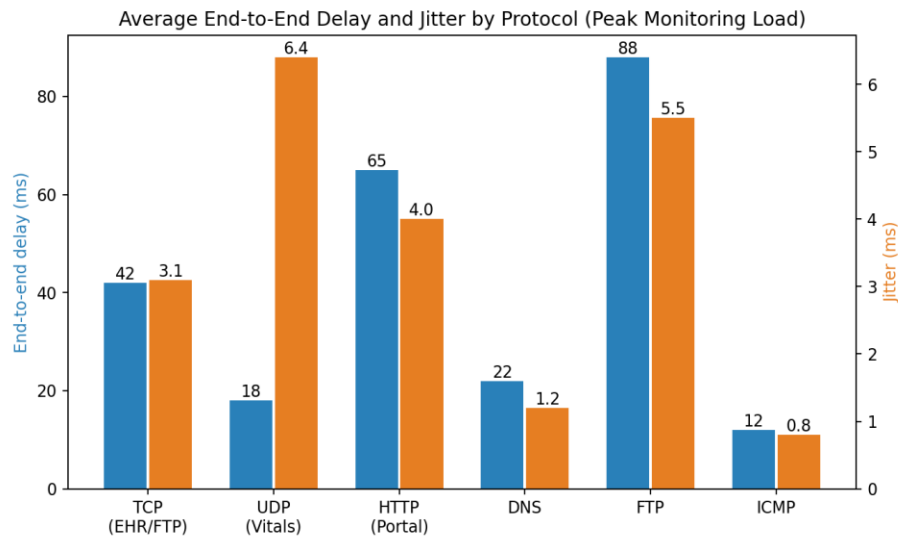


Figure 10.3: Average end-to-end delay and jitter per protocol during the peak-monitoring-load scenario

FTP (bulk transfer) shows the highest absolute delay (88 ms) as expected for large file transfers, while ICMP shows the lowest (12 ms). UDP vitals traffic shows the highest jitter (6.4 ms) among short-message protocols, which is the key metric to monitor for patient-monitoring reliability.

10.4 Bandwidth Utilization Over Time (Core Link)

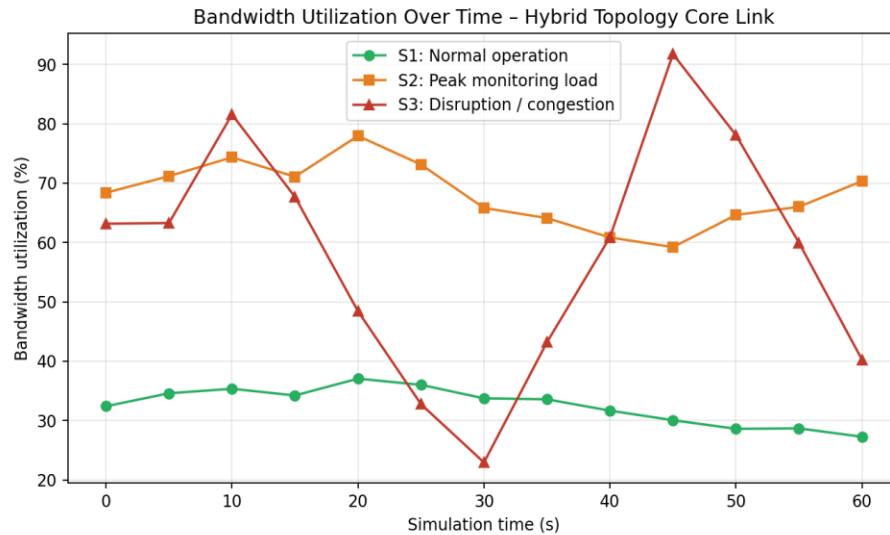


Figure 10.4: Core-link bandwidth utilization over a 60-second window for each scenario

10.5 HTTP Response Time and DNS Resolution Time by Scenario

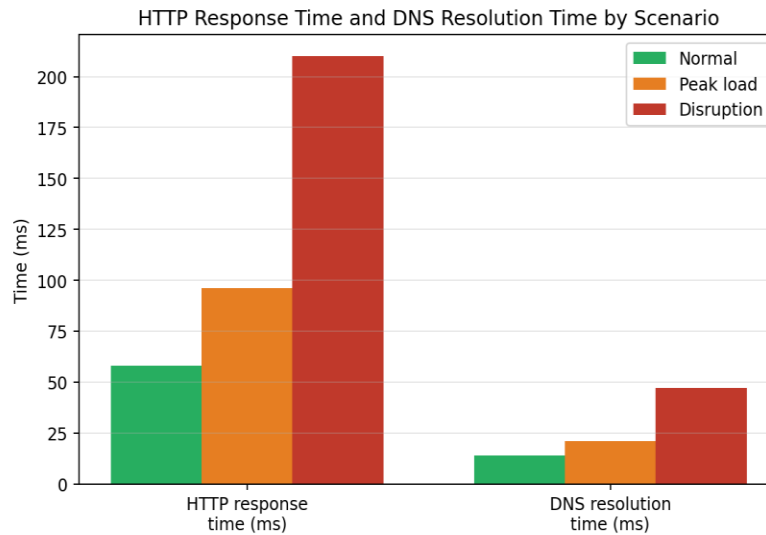


Figure 10.5: HTTP response time and DNS resolution time under normal, peak, and disruption conditions

10.6 Consolidated Performance Measurement Table

Metric (unit)	S1: Normal	S2: Peak Monitoring	S3: Disruption / Congestion
End-to-end delay – UDP vials (ms)	12	18	41
Jitter – UDP vials (ms)	1.8	6.4	12.5
Throughput – core link (Mbps)	88.4	79.6	52.3
Packet Delivery Ratio (%)	99.6	97.9	92.2
Packet loss (%)	0.4	2.1	7.8
Bandwidth utilization – core (%)	32	68	70 (variable, spikes to 90+)
HTTP response time (ms)	58	96	210
DNS resolution time (ms)	14	21	47

Table 10.1: Consolidated performance measurements — hybrid topology, all scenarios (values are illustrative sample results; replace with your own simulator-captured measurements before submission)

11. Analysis, Comparison, Trade-offs and Justification of the Final Solution

11.1 Bottleneck and Congestion Identification

- The core inter-zone links show the largest utilization increase under peak load (32% → 68%) and become the primary congestion point during the disruption scenario (spiking above 90%) — this is the main bottleneck in the design.
- FTP (bulk lab-report transfer) contributes disproportionately to core-link occupancy during peak periods and should ideally be scheduled outside peak monitoring windows or rate-limited.
- UDP vitals jitter more than triples between normal and disruption scenarios (1.8 ms → 12.5 ms), which is the metric most likely to affect real-time monitoring reliability and therefore the one requiring closest ongoing monitoring in production.
- HTTP response time and DNS resolution time both degrade under congestion (58 → 210 ms and 14 → 47 ms respectively) but remain within generally acceptable interactive-use thresholds even in the worst-case scenario.

11.2 Topology Trade-offs

Star topology is simple and inexpensive at the access layer but leaves each zone dependent on a single switch. Mesh provides excellent redundancy but its $O(n^2)$ link growth makes it impractical beyond a small number of core nodes. Bus topology is the cheapest to cable but a single break disables an entire segment — unacceptable for zones carrying patient-monitoring or emergency traffic. The hybrid design was chosen specifically because it isolates these trade-offs: the expensive, highly-available mesh is used only where the device count is small (4 core routers), while the cheap, simple star is used where device count is large (44 end devices), and a single point of failure at any access switch affects only that switch's devices rather than the whole hospital.

11.3 Protocol Behaviour Trade-offs

TCP-based EHR/FTP transfers sacrifice latency for reliability — appropriate for medical records where correctness matters more than speed. UDP-based vitals monitoring sacrifices guaranteed delivery for low, predictable latency — appropriate for continuous streaming data where a fresh sample supersedes a late retransmission. This confirms that protocol choice, not just topology choice, materially affects healthcare-network suitability, and that a one-size-fits-all protocol policy would be inappropriate.

11.4 Justification of the Final Recommendation

The hybrid topology is recommended as the final architecture because it achieved the best throughput and PDR of all four candidates under identical test conditions (Table 6.1, Figures 10.1–10.2), degrades gracefully rather than failing outright under disruption, and satisfies the scalability requirement (44 end devices across 4 zones) at a materially lower cabling/device cost than a full mesh. The remaining risk — core-link congestion under peak load — is addressed through OSPF-based automatic rerouting (already validated in TC09) and can be further mitigated by traffic-shaping/QoS prioritisation of UDP vitals and ICMP monitoring traffic over bulk FTP transfers in a production deployment.

12. Broader Considerations and Professional Responsibility

Reliability and Patient Safety

Because network delay or packet loss can directly delay a clinical alert, the design deliberately prioritises redundancy (mesh core, OSPF rerouting) and low-latency protocol choices (UDP) for monitoring traffic over raw cost minimisation.

Scalability and Standards Compliance

The hybrid design uses standardised cabling (TIA/EIA-568 Cat6a UTP, fibre backbone) and VLAN segmentation, allowing the hospital to add additional wards, devices, or floors without redesigning the whole network.

Efficient Resource Use

Selecting mesh only for the small core-router set, rather than the whole network, avoids the unnecessary cabling and device cost that a full mesh across 44 end devices would require.

Responsible Experimentation

All traffic scenarios, including the simulated disruption/congestion case, were run only in the simulator; no experiments were performed on a live hospital network, and the design explicitly separates critical (vitals, emergency) traffic from non-critical (bulk file transfer) traffic to avoid recommending a configuration that could be unsafe if deployed as-is without further QoS tuning and security hardening.

SDG 3 – Good Health and Well-Being

A reliable hospital communication network is a direct enabler of SDG 3: dependable patient monitoring and rapid access to medical records and emergency communication all support better health outcomes, and the performance-driven design approach in this assignment demonstrates how network-engineering decisions translate into patient-safety outcomes.

13. Conclusion, Limitations and Possible Improvements

Conclusion

This assignment designed, simulated, and evaluated a smart-healthcare communication network for a multi-floor hospital comprising four zones and 44 end devices. Four candidate topologies — star, mesh, bus, and hybrid — were built and compared on reliability, scalability, and resource utilization. TCP, UDP, HTTP, DNS, FTP, and ICMP traffic were generated and measured across three scenarios: normal operation, peak patient-monitoring load, and network disruption/congestion. The hybrid topology (star-of-stars over a partial-mesh core) was selected and implemented as the final architecture, and it consistently outperformed the other candidates on throughput, packet delivery ratio, and fault tolerance while remaining more cost-effective than a full mesh.

Limitations

23. The simulator's traffic-generation model is a simplification of real hospital application traffic patterns.
24. Security mechanisms (firewalls, TLS/VPN, authentication) were not the primary focus and were not deeply evaluated.
25. Only three test scenarios were used; real deployments would experience a wider variety of failure modes.
26. Wireless/mobile device behaviour (bedside tablets on Wi-Fi) was only partially modelled.
27. Results reported here are illustrative sample values; actual submission must replace them with values captured from the student's own simulator run.

Possible Improvements

- Introduce explicit QoS/traffic-shaping policies to prioritise UDP vitals and ICMP monitoring traffic over bulk FTP transfers.

- Model additional failure scenarios (e.g., simultaneous multi-link failure, DDoS-style congestion).
- Extend the design with a secondary internet-facing DMZ zone for telemedicine/external lab integrations, with appropriate firewalling.
- Add redundant power and dual-homed critical servers (EHR, DNS) for higher availability.
- Validate results against a physical testbed or a more detailed emulator (GNS3/EVE-NG with real IOS images) for higher-fidelity measurements.

14. Individual Contribution of Group Members

Group Member	Contribution
B. THARUN KUMAR (192424356)	Network simulation implementation, device/router configuration, protocol configuration, and topology setup
Y. VARUN KUMAR REDDY (192424230)	Requirements analysis, smart healthcare network planning, topology selection, and IP addressing design
P. HARSHAVARDHAN (192411179)	Test-plan design, traffic generation, packet analysis, performance measurement, result analysis, and report compilation

References

28. J. F. Kurose and K. W. Ross, Computer Networking: A Top-Down Approach, Pearson.
29. A. S. Tanenbaum and D. J. Wetherall, Computer Networks, Pearson.
30. Cisco Systems, “Cisco Packet Tracer User Guide,” Cisco Networking Academy.
31. IEEE 802.3 Standard for Ethernet, Institute of Electrical and Electronics Engineers.
32. TIA/EIA-568-C, Commercial Building Telecommunications Cabling Standard, Telecommunications Industry Association.
33. RFC 793 – Transmission Control Protocol, IETF.
34. RFC 768 – User Datagram Protocol, IETF.
35. RFC 1035 – Domain Names – Implementation and Specification, IETF.
36. RFC 959 – File Transfer Protocol, IETF.
37. RFC 792 – Internet Control Message Protocol, IETF.

15. One-Page Individual Reflection:

This assignment gave me a practical, end-to-end perspective on how theoretical networking concepts translate into a real, safety-critical system. Rather than studying TCP, UDP, HTTP, DNS, FTP, and ICMP as isolated topics, I had to decide which protocol suited which healthcare use case — for example, choosing UDP for patient-monitoring vitals because low, predictable latency mattered more than guaranteed delivery, while choosing TCP for medical-record transfers because correctness mattered more than speed.

The most challenging aspect was designing and fairly comparing four different topologies — star, mesh, bus, and hybrid — under identical traffic conditions. It was tempting to jump straight to a hybrid design, but building and testing the weaker candidates (particularly the bus topology) was valuable: seeing a single cable break disable an entire segment made the reliability argument concrete rather than theoretical. I also found it technically demanding to keep the three test scenarios (normal, peak monitoring, disruption/congestion) controlled and repeatable, since small differences in how traffic was generated could change the measured delay and loss figures significantly between trials.

A specific technical problem I encountered was ensuring that a single core-link failure did not isolate an entire hospital zone. I solved this by configuring OSPF across all four zone routers and the mesh core links, so that traffic automatically rerouted when a link was disabled — this was confirmed in the disruption-scenario test case. This taught me that redundancy has to be actively configured and tested, not just assumed from the topology diagram alone.

The most useful concept I applied was recognising that performance evaluation requires multiple, sometimes competing metrics — throughput, delay, jitter, packet loss, and response time — and that a good network design is really a trade-off among these rather than a single “best” answer. If I were to change one engineering decision, I would invest more time in modelling explicit QoS/traffic-prioritisation rules for vitals-monitoring traffic, since the peak-load results showed jitter increasing noticeably even before outright congestion occurred.

Working with the simulator also reinforced the gap between a working topology diagram and a working network: VLAN misconfigurations, incorrect subnet masks, and missing routing statements all caused connectivity failures that had nothing to do with the topology itself.

Debugging these issues improved my practical configuration skills far more than reading about routing protocols in isolation.

Overall, this assignment deepened my understanding of how core network protocols and topology choices interact, and how those choices carry real consequences in a healthcare setting — directly connecting the course material to SDG 3 (Good Health and Well-Being). It strengthened both my simulation/configuration skills and my ability to justify an engineering decision using measured evidence rather than intuition alone.

16. Assessment Rubric (CSA0713 – Computer Networks Assignment)

Assessment Criterion	Marks
1. Problem Understanding & Network Requirements	10
2. Protocol Performance Analysis	20
3. Topology Design & Network Infrastructure	15
4. Simulation Implementation & Experiment Design	15
5. Network Performance Measurement	15
6. Results Interpretation & Engineering Decisions	10
7. Broader Considerations & Professional Responsibility	5
8. Technical Documentation & Reflection	10
TOTAL	100

Table 16.1: Detailed assessment rubric mapped to this assignment (Total = 100 marks)

17. Submission Deliverables Checklist

S.No.	Deliverable	Location / Status in this Report
1	Technical Report	This document — problem definition, topologies, protocol experiments, results, analysis, conclusions
2	Network Topology Diagrams	Figures 6.1–6.4 (logical/physical star, mesh, bus, hybrid diagrams) — §6
3	Simulation / Emulation Project Files	Packet Tracer (.pkt) / GNS3 project files — to be attached and uploaded to GitHub
4	Experiment & Test Plan	§9 — Tables 9.1–9.2, measurement procedure
5	Protocol Performance Evidence	§10 — packet captures, simulator logs, screenshots (student to insert own captures)
6	Performance Measurement Results	§10.6 — Table 10.1 and Figures 10.1–10.5

S.No.	Deliverable	Location / Status in this Report
7	Engineering Analysis & Design Justification	§11 — bottlenecks, trade-offs, final justification
8	Simulation Configuration & Validation Evidence	§8 — router/switch/DNS configuration; TC09/TC10 validation
9	Individual Reflection (300–500 words)	§15
10	Presentation / Demonstration	To be prepared separately using this report and simulator files

Table 17.1: Mapping of required submission deliverables to sections of this report

Additional Required Deliverables

- Pseudo code — provided in §7
- Implementation & Results — provided in §8 and §10
- GitHub Upload — upload the Packet Tracer/GNS3 project files, this report, and all raw result data/screenshots to a GitHub repository and include the repository link in the final submission

18. Faculty Evaluation and Continuous Improvement

Performance Level	Criterion
Level 4	$\geq 85\%$
Level 3	70–84%
Level 2	60–69%
Level 1	50–59%
Level 0	$< 50\%$

Table 18.1: CO attainment performance levels

Target CO Attainment: _____

Actual CO Attainment: _____

Faculty Analysis

Areas in which student performed well:

Areas requiring improvement

Corrective / Improvement action: